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Department of Information Technology

End Term Examination – November 2025

Subject Code:- IT-3003

Subject Name:- Signal and Communication Systems

Max Marks:- 60

Time:- 03 Hrs.

Note:

1. All the questions are compulsory and while solving write question number and its part clearly.
2. All questions carry equal marks.
3. Solve questions in given sequence and all the sub parts should be at same place.
4. Draw neat and clean sketches.

S. No.	Questions	Marks
1.	a) Evaluate and sketch the output $y(t)=x(t)*h(t)$, where $*$ denotes convolution operator. The input signal and the impulse response are given as: $x(t)=u(t)-u(t-3)$ and $h(t)=u(t)-u(t-2)$, where $u(t)$ is the unit step function.	6
	b) Find the Laplace transform $X(s)$ and sketch the ROC for the following signals $x(t)$: i. $x(t)=e^{-2t}u(t)+e^{-3t}u(t)$ ii. $x(t)=e^{-3t}u(t)+e^{2t}u(-t)$ iii. $x(t)=e^{2t}u(t)+e^{-3t}u(-t)$	6
2.	a) Find the Fourier transform $X(\omega)$ for the following signals $x(t)$: i. $x(t)=u(t+0.5)-u(t-0.5)$ ii. $x(t)=e^{-a t }$, $a>0$ iii. $x(t)=\frac{1}{x^2+t^2}$, $a>0$	6
	b) Draw the block diagram of the Communication System and explain the function of each block..	6
3.	a) Why audio signals cannot be transmitted directly? Compare analog and digital modulation in terms of bandwidth and noise. Why is the modulation index important in Amplitude Modulation?	6
	b) Derive the expression for the instantaneous voltage of amplitude modulated wave. What are the advantages and disadvantages of Single Side Band over Double Side Band? Explain Single Side Band modulation.	6
4.	a) FM signal is represented as $v(t)=10 \sin (2\pi \times 10^6 t + 5 \sin (2\pi \times 10^3 t))$. Find: (i) The frequency deviation (ii) Modulation index (iii) Bandwidth using Carson's rule.	6

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	b) Explain working principle of Pulse Code Modulation transmitter and receiver. Also brief about quantization noise. Write technical note on Pulse Width Modulation and Pulse Position Modulation.	6
5.	a) Justify the significance of Sampling Theorem. An Frequency Shift Keying system uses frequency deviation of ± 5 kHz with a bit rate of 2 kbps. Estimate its bandwidth using Carson's rule.	6
	b) Explain following terms briefly : i) Tuned Circuit Frequency Modulation Detector ii) Amplitude Shift Keying, Frequency Shift Keying, Phase Shift Keying waveforms for input 1100100101.	6

$$(1 + \frac{P}{A}) \times \cancel{2} + 2\Delta f$$